Integrating Semisupervision and Prototype Contrastive Learning for Robust Semantic Change Detection in Remote Sensing Images

Yupeng Gao, Tianyu Li, Keli Wang, Guoqing Wang, *Member, IEEE*, Jiayi Ma, *Senior Member, IEEE*, Ningjuan Ruan, and Yang Yang, *Senior Member, IEEE*

***Abstract*—The advancement of semantic change detection (SCD) in remote sensing imagery has been significantly limited by the scarcity of high-quality annotated datasets, as acquiring accurate multitemporal labels is both labor-intensive and reliant on domain-specific expertise. To address this challenge, we propose a novel framework that integrates semi-supervised learning with prototype contrastive learning for robust SCD in remote sensing images. We decompose the complex change detection problem into complementary learning tasks: semi-supervised learning guided by adaptive thresholds and feature representation enhancement driven by prototype similarity. Specifically, we introduce three key innovations: first, we propose an adaptive threshold strategy that dynamically adjusts confidence thresholds based on class distribution imbalances, effectively expanding training samples for rare categories; second, we design a prototype-guided contrastive learning (PGCL) module with memory banks and multilayer feature fusion, which reshapes the feature space to cluster semantically similar features while separating different ones, particularly improving the model's ability to distinguish hard-to-classify classes; Finally, a semantic-change consistency regularization (SCCR) is introduced to enforce coherence between semantic segmentation and change detection outputs, ensuring that predicted changes align logically with the underlying semantic transitions. Extensive experiments on three public benchmark datasets, SECOND, Landsat-SCD, and HiU-CD mini, demonstrate that SemiSCD outperforms state-of-the-art methods.**

***Index Terms*—Contrastive learning, prototype, semantic change detection (SCD), semisupervision.**

Received 21 October 2025; revised 1 December 2025; accepted 15 December 2025. Date of publication 18 December 2025; date of current version 8 January 2026. This work was supported in part by the National Natural Science Foundation of China under Grant U23B2011, Grant U20B2063, and Grant 62220106008; in part by the Key Research and Development Program of Zhejiang under Grant 2024SSYS0091; in part by the Sichuan Science and Technology Program under Grant 2024NSFTD0034; and in part by the Beijing Key Laboratory Fund under Grant 202410FKY00390. *(Corresponding authors: Guoqing Wang; Tianyu Li.)*

Yupeng Gao, Tianyu Li, Keli Wang, Guoqing Wang, and Yang Yang are with the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu 610054, China (e-mail: 202411081510@std.uestc.edu.cn; cosmos.yu@hotmail.com; 202421080426@std.uestc.edu.cn; gqwang0420@hotmail.com; yang.yang@uestc.edu.cn).

Jiayi Ma is with the School of Electronic Information, Wuhan University, Wuhan 430072, China (e-mail: jiayima@whu.edu.cn).

Ningjuan Ruan is with Beijing Institute of Space Mechanics and Electricity, Beijing 100094, China (e-mail: ruanningjuan@163.com).

Digital Object Identifier 10.1109/TGRS.2025.3645461

I. Introduction

REMOTE sensing semantic change detection (SCD) is a fundamental task in Earth observation data analysis, which not only identifies "where changes occur" but also determines "what has changed" [1], [2], [3]. This provides crucial spatiotemporal information that supports applications in environmental monitoring [4], [5], urban planning [6], [7], and disaster assessment [8], [9].

Traditional methods such as pixel differencing and principal component analysis were applied in early investigations [12], while in recent years, deep learning-driven SCD methodologies have flourished along three principal technical routes: postclassification change detection (PCD) approaches conceptualize change identification as a postprocessing stage of semantic segmentation, effectively leveraging pretrained model knowledge for multitemporal image analysis [13]; direct comparison (DC) methodologies analyze spatial features through encoders with weight sharing, maintaining feature consistency throughout the change analysis process [14]; and multitask learning architectures endeavor to optimize semantic segmentation and change detection as parallel tasks simultaneously, achieving information complementarity and joint optimization [15], [16].

Despite significant progress on standard benchmarks, existing methods still face three fundamental challenges when applied to complex real-world remote sensing scenarios (as shown in Fig. 1). First, annotations in remote sensing datasets exhibit significant structural scarcity—typically, only pixels within change regions are precisely labeled, while the vast majority of pixels in nonchange areas remain unlabeled. This annotation structure leads to substantial wastage of potentially informative data and fundamentally limits the model's ability to learn comprehensive semantic representations from the entire scene. Second, the distribution of land cover classes in remote sensing scenes is highly imbalanced, with rare but critical categories (e.g., water bodies and playgrounds) often comprising less than 1% of the total pixels, making it extremely difficult for models to achieve balanced performance across all classes. Third, bitemporal remote sensing images present complex change patterns where models must precisely distinguish between mere appearance changes (e.g., seasonal variations and illumination differences) and genuine semantic

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

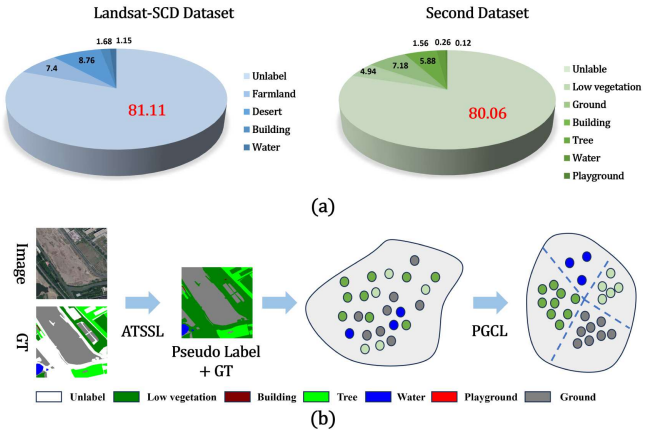


Fig. 1. Class distribution in remote sensing SCD datasets and semi-supervised method logic. (a) Class imbalance phenomenon in Landsat-SCD [10] and SECOND dataset [11] is shown, where gray areas represent *unlabel pixels within each image* (unchanged regions without semantic annotations) accounting for 81.11% and 80.06% of total pixels, respectively. (b) Semi-supervised methods generate pixel-level pseudo-labels for these unlabel pixels within remote sensing images.

transitions, posing severe challenges to feature representation learning [17], [18]. Moreover, practical deployment of SCD systems requires robustness against image quality variations commonly encountered in operational scenarios, such as atmospheric noise and sensor artifacts, which can significantly degrade model performance if not properly addressed.

To address these challenges, this article introduces SemiSCD, a novel framework that integrates semi-supervised learning with prototype contrastive learning for robust SCD in remote sensing images. As shown in Fig. 1(b), our method begins by obtaining initial feature representations through supervised learning and then simultaneously optimizes the feature space in two ways: in one way, it employs an adaptive threshold mechanism to mine high-quality pseudo-labels from unlabeled data, thereby expanding the training samples. In another way, it reconstructs the feature space distribution using prototype similarity-guided feature contrastive learning, which fosters the clustering of features with similar semantics and the separation of features with different semantics, particularly enhancing the discriminative ability for categories that are easily confused. Additionally, we establish semantic-change consistency constraints between semantic segmentation and change detection to ensure logical coordination in predictions. The main contributions of this article are summarized as follows.

1) A semi-supervised learning framework, SemiSCD, is specifically designed for remote sensing SCD. It effectively utilizes limited labeled data and large amounts of unlabeled data, significantly improving model performance in complex scenarios.
2) We propose an adaptive threshold semi-supervised learning (ATSSL) mechanism to address class imbalance in pseudo-label generation and a prototype similarity-guided contrastive learning method to capture class similarity in feature space. Additionally, we introduce semantic-change consistency regularization (SCCR) to ensure logical coordination across task predictions.
3) Extensive experiments on two public datasets, SECOND and Landsat-SCD, demonstrate that our proposed method outperforms state-of-the-art approaches.

II. Related Work

A. SCD in Remote Sensing

SCD has evolved significantly with the advancement of deep learning. Early approaches relied on pixel-level differencing and postclassification comparison [19], but recent methods have embraced end-to-end deep learning architectures along three main technical routes.

Postclassification methods treat change detection as a downstream task of semantic segmentation, leveraging pretrained models to analyze multitemporal imagery independently before comparing results [20]. While conceptually straightforward, these methods suffer from error propagation and fail to capture temporal dependencies effectively.

DC methods extract features from bitemporal images through shared-weight encoders and identify changes by comparing spatial features directly [14]. Tang et al. [21] proposed the ClearSCD model, which comprehensively leverages semantic information and change relationships through an end-to-end framework, significantly enhancing SCD performance in high-resolution remote sensing imagery. Chang et al. [22] introduced a triple-branch hybrid attention network with bitemporal feature joint refinement strategies, improving sensitivity to subtle changes and robustness to noise.

Multitask learning methods jointly optimize semantic segmentation and change detection as parallel tasks [23], [24]. Li et al. [25] enhanced precise boundary localization through decoder optimization. Chen et al. [26] proposed the ChangeMamba model, which pioneered the application of state-space models (SSMs) to remote sensing change detection. Zhou et al. [27] developed SChanger, which examines change detection from the dual perspectives of semantic change and spatial consistency, designing an adaptive feature interaction mechanism. Long et al. [28] introduced a hierarchical semantic graph interaction network for change detection, enhancing the model's understanding of contextual dependencies.

The emergence of foundation models has introduced new paradigms to the remote sensing change detection field. Sun et al. [29] explored the application of the segment anything model [30] as a semantic knowledge extractor for change detection. However, a fundamental limitation persists across all these approaches: they rely on fully supervised learning paradigms and cannot effectively leverage the vast amount of unlabeled pixels in remote sensing datasets, where typically over 80% of pixels remain without annotations. Moreover, extreme class imbalance remains an unresolved challenge, particularly for rare but critical categories such as water bodies and playgrounds.

B. Semi-Supervised Learning for Remote Sensing

The annotation bottleneck in remote sensing has motivated extensive research in semi-supervised learning, which

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

effectively reduces the cost barrier of data annotation by simultaneously utilizing both labeled and unlabeled data. The core challenge lies in generating reliable pseudo-labels from unlabeled data while preventing error accumulation.

Pseudo-labeling strategies form the foundation of modern semi-supervised methods. FixMatch [31] established the paradigm of combining consistency regularization with confidence-based thresholding, demonstrating that high-confidence predictions on weakly augmented unlabeled images can serve as effective supervision for strongly augmented versions. Mai et al. [32] proposed RankMatch, which optimized consistency regularization strategies by ranking prediction confidence and dynamically adjusting pseudo-label selection. Wang et al. [33] improved model stability by introducing a density-descending feature perturbation method, which gradually increases perturbation strength based on feature density to enhance robustness. For remote sensing specifically, Zuo et al. [34] developed an instance-level semi-supervised framework to enhance complex scene recognition capabilities. Hoyer et al. [35] integrated vision-language guided learning approaches in their SemiVL framework, leveraging textual descriptions to improve semantic understanding.

Teacher–student architectures with exponential moving average (EMA) [36] have become standard practice in semi-supervised learning. The teacher model generates pseudo-labels while the student learns from both labeled and pseudo-labeled data. However, existing methods typically apply uniform confidence thresholds across all categories, failing to account for the dramatic class frequency imbalance inherent in remote sensing scenes. For example, in the SECOND dataset, water comprises only 0.26% of pixels while buildings account for over 5%. This leads to systematic undersampling of rare but important categories, as fixed thresholds favor majority classes with more training examples and higher prediction confidence.

C. Contrastive Learning for Feature Representation

Contrastive learning, as a key technology for feature representation optimization, has achieved significant advances in remote sensing fields. The core principle is to learn discriminative representations by pulling positive pairs together while pushing negative pairs apart in the embedding space.

General contrastive frameworks such as SimCLR [37] and MoCo [38] have demonstrated remarkable success in self-supervised learning. For semantic segmentation, pixel-level contrastive learning [39] enhances feature discrimination by treating pixels of the same class as positives and different classes as negatives. However, these methods typically operate on final-layer features only, neglecting complementary information from intermediate layers that capture both high-level semantics and fine-grained spatial details.

Addressing class imbalance through contrastive learning has attracted increasing attention. Xie et al. [40] enhanced representation capabilities through feature distribution uncertainty modeling, which accounts for intraclass variance when constructing contrastive pairs. Xin et al. [41] combined dual-teacher networks with biased contrastive learning to specifically address class imbalance issues, adjusting sampling probabilities to emphasize minority classes. However, these approaches rely on heuristic rebalancing strategies rather than explicitly modeling class similarity relationships.

Prototype-based contrastive learning offers a principled approach to capturing class-level semantics. Liao et al. [42] improved model cross-domain transferability with multiprototype contrastive learning, demonstrating that maintaining multiple prototypes per class can capture intraclass diversity. Zhang et al. [43] effectively resolved data and model heterogeneity challenges with their FedTGP framework, using trainable global prototypes to aggregate knowledge across distributed clients. Yet, existing prototype methods do not leverage similarity-guided sampling strategies. Intuitively, sampling hard negatives from the most similar classes and diverse positives from within-class variations should be more effective than random sampling for learning discriminative features, but this insight remains unexplored in remote sensing applications.

Memory mechanisms for storing and aggregating historical features have shown promise in continual learning [44] and video understanding [45]. However, their application to remote sensing change detection—where maintaining class prototypes across diverse geographic scenes could provide robust representations—remains uninvestigated. Moreover, the integration of contrastive learning with semi-supervised frameworks specifically for SCD has not been explored. Building upon these foundations, this work introduces a novel integration of ATSSL and prototype-guided contrastive learning (PGCL) specifically tailored for remote sensing SCD.

III. Method

A. Overall Framework

The framework of our proposed SemiSCD is presented in Fig. 2. Given two remote sensing images from different time phases $I_1, I_2 \in \mathbb{R}^{H\times W\times C}$, where H, W, and C represent the image height, width, and the number of channels, respectively, SemiSCD aims to accomplish two primary tasks: generating semantic segmentation maps for both temporal phases ($S_1, S_2 \in \mathbb{R}^{H\times W\times K}$, where K represents the number of semantic categories) and producing a binary change map ($M \in \mathbb{R}^{H\times W\times 1}$) to identify changed regions. As shown in Fig. 2, our network architecture consists of weight-shared encoder Φ_{encoder}, semantic segmentation decoder $\Phi_{\text{decoder,SS}}$, and a change detection decoder $\Phi_{\text{decoder,CD}}$. In the feature extraction stage, HRNet40 [46] is employed as the backbone feature extractor for both temporal images, due to its superior ability to maintain high-resolution representations throughout the network, which is crucial for pixel-level change detection tasks. For the decoder structure, we utilize a feature pyramid network (FPN) [47] to effectively process multiscale features. The two semantic segmentation decoders process features from each temporal phase separately

$$S_1 = \Phi_{\text{decoder,SS}}(F_1); \quad S_2 = \Phi_{\text{decoder,SS}}(F_2). \tag{1}$$

The change detection decoder receives encoder features and semantic segmentation features from both temporal phases to detect changed regions. Specifically, it first computes the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

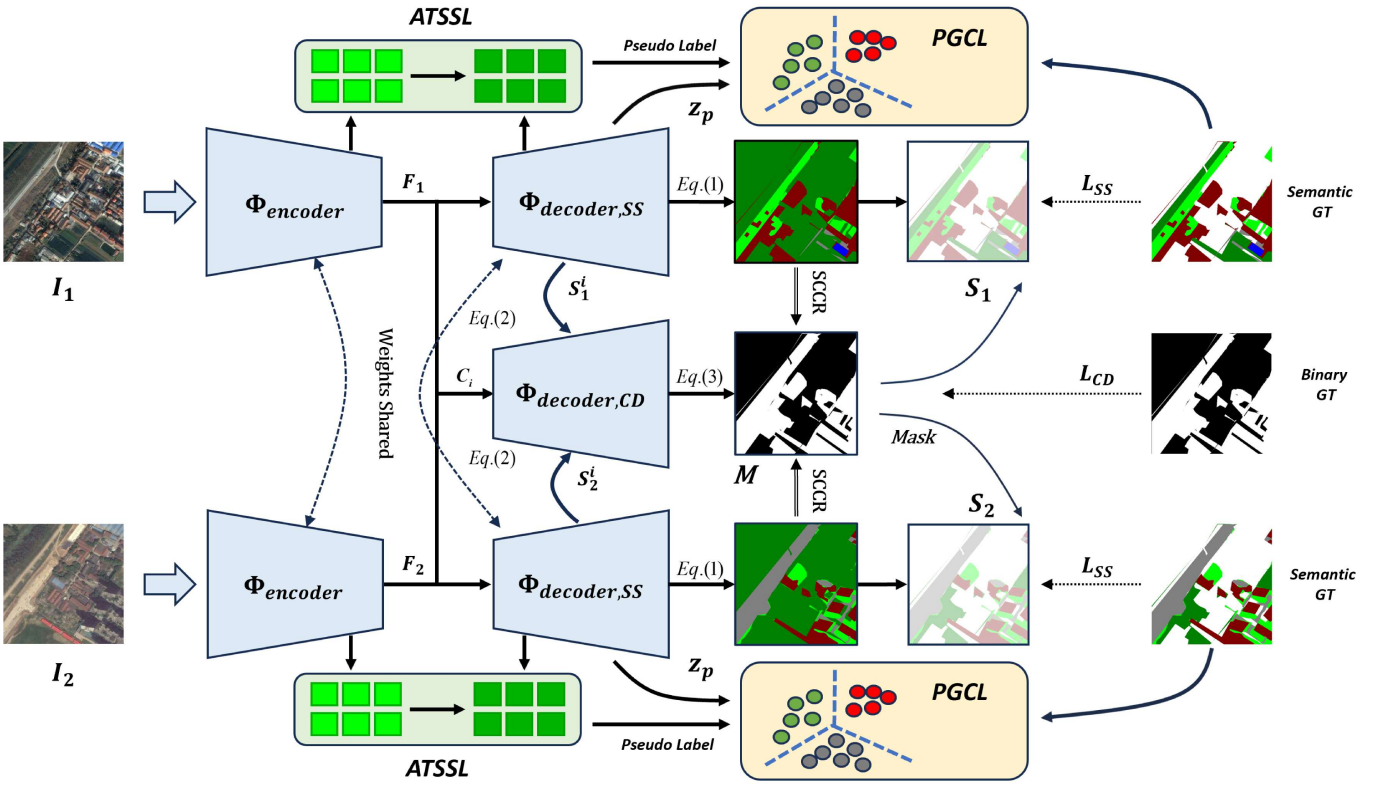


Fig. 2. Overall architecture of the SemiSCD framework with semi-supervised learning strategy. Training proceeds in two stages: Stage 1 employs supervised learning on labeled pixels to establish reliable feature representations, and Stage 2 activates semi-supervised learning where the teacher model generates pseudo-labels via the ATSSL module and the student model learns from both labeled and pseudo-labeled data. The PGCL module enhances feature discrimination using prototypes, while the SCCR module ensures prediction consistency.

absolute differences of encoder features $C_i = |F_1^i - F_2^i|$ from different hierarchical levels, where $i = 1, 2, 3$, and 4 represents the four different scale levels of the feature pyramid, and the absolute differences of semantic features from the segmentation decoders $S_{\text{diff}}^i = |S_1^i - S_2^i|$. For processing encoder feature differences, the feature differences C_i are first elementwise multiplied with the corresponding semantic difference features S_{diff}^i at each layer

$$O_i = C_i \odot S_{\text{diff}}^i. \tag{2}$$

Then, O_i undergoes layer-by-layer upsampling and fusion to obtain the final feature O_{final}, where $\odot$ represents the elementwise multiplication operation. Finally, O_{final} passes through a classifier to obtain the binary change map

$$B = \Phi_{\text{decoder,CD}}\left(O_{\text{final}}\right). \tag{3}$$

Semantic change labels are derived by masking the semantic segmentation results with the binary change map to focus only on changed regions.

To fully utilize unlabeled pixel data and address class imbalance, we design an ATSSL framework based on a teacher–student model. The teacher model updates its parameters through EMA mechanism

$$\theta_{\text{teacher}} = \alpha \cdot \theta_{\text{teacher}} + (1 - \alpha) \cdot \theta_{\text{student}} \tag{4}$$

where $\alpha = 0.99$ and generates pseudo-labels for unlabeled data, while the student model is trained jointly using supervised data and pseudo-labels. Additionally, we propose a PGCL framework that incorporates prototype memory banks (PMBs) and similarity-based sampling strategies to enhance discrimination capability for easily confused categories. Finally, we introduce an SCCR mechanism that minimizes discrepancies between change detection and semantic change probabilities.

B. Adaptive Threshold Semi-Supervised Learning

1) Clarification on Semi-Supervised Setting: Unlike traditional semi-supervised learning that partitions the dataset into labeled and unlabeled subsets, our semi-supervised approach leverages the inherent annotation structure of SCD datasets. Specifically, in SCD datasets, all training images are used, but within each image pair, only pixels in changed regions have semantic labels, while pixels in unchanged regions lack semantic annotations (as semantic categories do not change in these areas and thus are not annotated). This results in approximately 80% of pixels being unlabeled across the dataset [Fig. 1(a)]. Our semi-supervised learning framework exploits these unlabeled pixels within each training image through pseudo-labeling, rather than excluding entire images from supervision. This setting ensures fair comparison with fully supervised baselines, as all methods use the same training images—the difference lies in whether unlabeled pixels within images are utilized.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

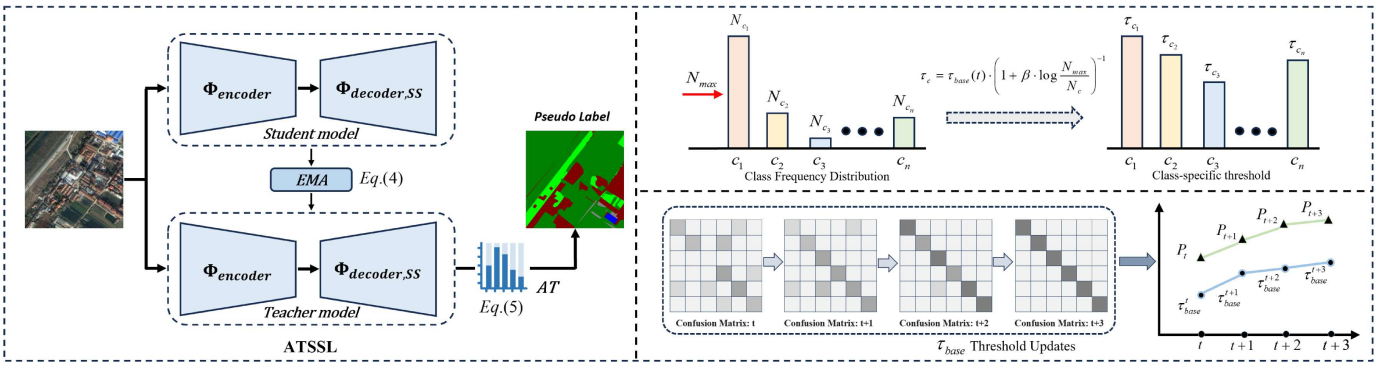


Fig. 3. Detailed illustration of the ATSSL mechanism in Stage 2. (Left) Semi-supervised training pipeline in Stage 2, showing how the teacher model (updated via EMA) generates pseudo-labels for unlabeled pixels, which are then combined with labeled data to train the student model. (Top right) Transformation from class frequency distribution to class-specific confidence thresholds. (Bottom right) Adaptive threshold mechanism based on classification performance.

2) Problem Formulation: Given a dataset $\mathcal{D} = \{(I_1^i, I_2^i, M_1^i, M_2^i)\}_{i=1}^N$, where I_1^i and I_2^i represent the bitemporal images and M_1^i and M_2^i denote the pixel-level semantic segmentation annotation masks containing both labeled pixels (with semantic class labels) and unlabeled pixels (without annotations) for each temporal phase, our objective is to jointly train a model using the sparse labeled pixels and abundant unlabeled pixels within each image pair to enhance performance on the test set.

3) Two-Stage Training Strategy: To ensure the quality of pseudo-labels and prevent error accumulation in semi-supervised learning, we adopt a two-stage training strategy. In Stage 1, the model is trained with fully supervised learning on labeled pixels only to establish stable and reliable feature representations. This supervised pretraining phase is crucial for the teacher model to generate high-quality pseudo-labels, as unreliable pseudo-labels in early training can mislead the model and degrade performance [31]. At the end of Stage 1, we initialize the teacher model by copying the trained student model parameters: $\theta_{\text{teacher}} \leftarrow \theta_{\text{student}}$. The teacher model will subsequently be updated through EMA throughout Stage 2, as illustrated in Fig. 3, left. Once the model achieves sufficient performance on labeled data, we introduce the semi-supervised learning mechanism in Stage 2 to leverage unlabeled data while maintaining prediction reliability.

To address the extreme class imbalance challenge in remote sensing datasets, we propose an adaptive threshold strategy that dynamically adjusts confidence thresholds based on class frequency distribution. The overall mechanism is illustrated in Fig. 3, which shows the transformation from class frequency distribution to class-specific confidence thresholds, and the subsequent pseudo-label generation and model update process.

Class Statistics Precomputation: Before training begins, we compute class-specific statistics from the labeled pixel subset. For each semantic class $c \in \{1, \ldots, K\}$, we count the total number of labeled pixels N_c across all training images (shown in the "Class Frequency Distribution" panel of Fig. 3). We also record $N_{\max}$, the pixel count of the most frequent class among all labeled pixels. Importantly, N_c and $N_{\max}$ are computed once from the labeled pixel subset before training begins and remain fixed throughout both training stages. This design ensures that the adaptive thresholds are based on reliable ground-truth class distributions rather than potentially noisy estimates from unlabeled regions.

As shown in the top right of Fig. 3, the class frequency distribution is transformed into class-specific confidence thresholds through our adaptive threshold formula. For each class c, the pseudo-label generation threshold τ_c is calculated as

$$\tau_c = \tau_{\text{base}}(t) \cdot \left(1 + \beta \cdot \log \frac{N_{\max}}{N_c}\right)^{-1} \tag{5}$$

where $\tau_{\text{base}}(t)$ is the base threshold at epoch t (initialized as $\tau_{\text{base}}(0) = 0.6$), and $\beta = 0.1$ is the adjustment parameter. The resulting class-specific thresholds are visualized in the "Class-specific threshold" panel of Fig. 3, where rare classes receive lower thresholds (smaller τ_c) to increase their recall, while frequent classes maintain higher thresholds (larger τ_c) to ensure precision. By computing these statistics from labeled data only, our threshold adaptation mechanism provides stable guidance for pseudo-label generation throughout training.

4) Pseudo-Label Generation Process: During Stage 2, for each training iteration, the teacher model processes unlabeled pixels to generate class probability distributions. Specifically, for an unlabeled pixel p in nonchange regions, the teacher model produces a probability vector $\mathbf{p} = [p_1, p_2, \ldots, p_n]$, where n is the number of semantic classes (corresponding to classes $c_1, c_2, \ldots, c_n$ in Fig. 3). The predicted class $c_i = \arg\max_k p_k$ and its corresponding confidence score $p_{c_i} = \max_k p_k$ are compared against the class-specific threshold τ_{c_i} (where $\tau_{c_i} \in \{\tau_{c_1}, \tau_{c_2}, \tau_{c_3}, \ldots, \tau_{c_n}\}$ as shown in the "Class-specific threshold" panel of Fig. 3)

$$\text{Pseudo-label}(p) = \begin{cases} c_i, & \text{if } p_{c_i} \geq \tau_{c_i} \\ \text{discard}, & \text{otherwise.} \end{cases} \tag{6}$$

Only pixels whose prediction confidence exceeds their class-specific threshold are assigned pseudo-labels and included in the training loss; low-confidence predictions are discarded to prevent noise accumulation. This selective pseudo-labeling mechanism ensures that the student model is trained on high-quality pseudo-labels while adapting to class imbalance through differentiated threshold requirements.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

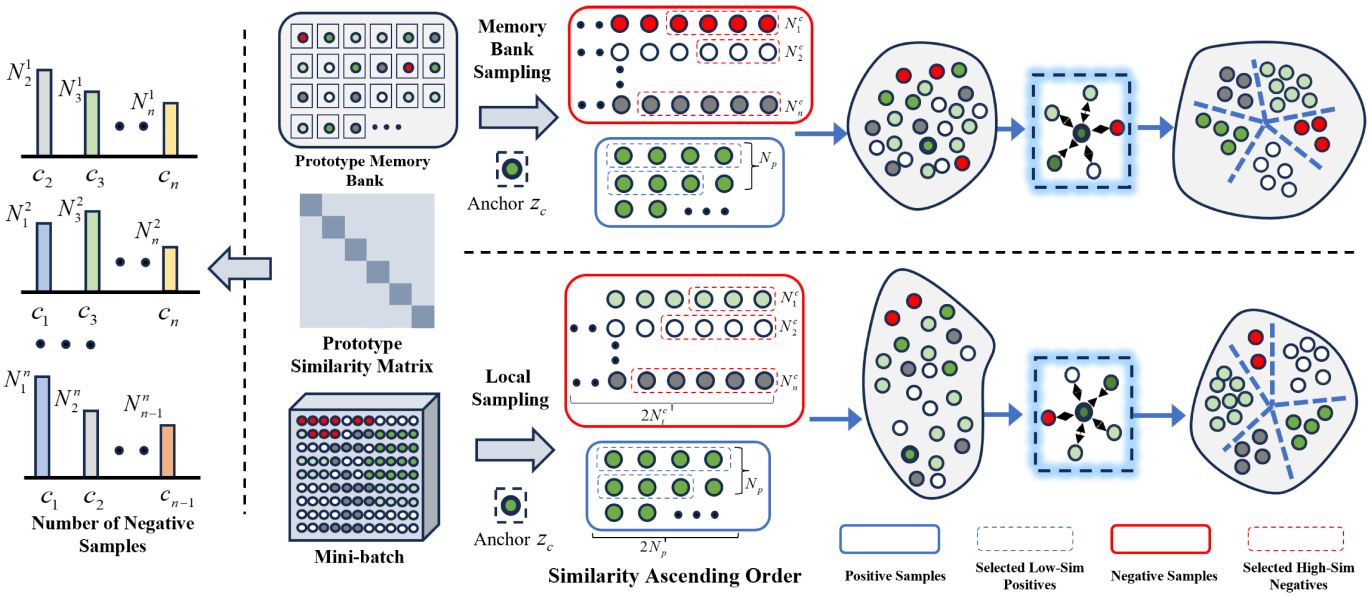


Fig. 4. PGCL module architecture. The diagram shows the memory bank sampling strategy, prototype similarity matrix construction, and the adaptive threshold mechanism that dynamically adjusts confidence thresholds based on class frequency distribution to address class imbalance in pseudo-label generation.

5) Dynamic Weight Adjustment and Loss Function: For unlabeled data, the pseudo-label weight λ^u_{sem} is dynamically adjusted based on pseudo-label reliability

$$\lambda^u_{\text{sem}} = \frac{N^p_r}{N^p_t} \tag{7}$$

where N^p_t is the total number of pseudo-label samples in the current batch (total number of pixels in nonchange regions), and N^p_r is the number of reliable pseudo-label samples that pass the filtering threshold. The overall semi-supervised loss function is defined as

$$\mathcal{L}_{\text{ss}} = \mathcal{L}_{\text{sup}} + \lambda^u_{\text{sem}} \cdot \mathcal{L}_{\text{pseudo}} \tag{8}$$

where $\mathcal{L}_{\text{sup}}$ and $\mathcal{L}_{\text{pseudo}}$ denote the cross-entropy loss on labeled pixels with ground-truth annotations and the cross-entropy loss on unlabeled pixels using pseudo-labels generated by the teacher model, respectively.

6) Base Threshold Adaptation: The base threshold $\tau_{\text{base}}(t)$ is dynamically updated at the end of each epoch based on validation performance

$$\tau_{\text{base}}(t) = \max\left(\tau_{\min}, \min\left(\tau_{\max}, \tau_{\text{base}}(t-1) + \gamma \cdot (P_t - P_{t-1})\right)\right) \tag{9}$$

where $\gamma = 0.1$, and the constraint conditions $\tau_{\min} = 0.4$ and $\tau_{\max} = 0.85$ are set empirically to prevent excessive threshold fluctuation. P_t represents the average pseudo-label accuracy on the validation set, calculated as

$$P_t = \frac{1}{|L|} \sum_{l \in L} \frac{\text{TP}^t_l}{\text{TP}^t_l + \text{FP}^t_l} \tag{10}$$

where L is the set of all classes, TP^t_l represents the number of samples consistently predicted as class l at both current and previous time points, and FP^t_l represents samples predicted as class l at the current time point but as other classes at the previous time point. This dynamic adjustment mechanism, illustrated by the confusion matrix evolution in Fig. 3, allows τ_{base} to adapt to model improvement: as pseudo-labels become more reliable (shown by increasing diagonal values in confusion matrices), the base threshold can be gradually adjusted to maintain high precision while maximizing the utilization of unlabeled data.

C. Prototype-Guided Contrastive Learning

The PGCL architecture illustrated in Fig. 4 incorporates PMBs and similarity-based sampling strategies to enhance discrimination capability for easily confused categories. Traditional contrastive learning typically uses final layer features as anchor representations, neglecting complementary information from different layers. We design a multilayer feature fusion anchor mechanism that combines high-level semantic information with intermediate spatial details. For a pixel p belonging to class c, its anchor feature z_a is

$$z_a = \alpha \cdot \text{proj}_L\left(F_L(p)\right) + (1-\alpha) \cdot \text{proj}_{L-1}\left(\frac{1}{|\Omega_c|} \sum_{q \in \Omega_c} F_{L-1}(q)\right) \tag{11}$$

where $F_L(p)$ and $F_{L-1}(q)$ are the features from final and penultimate layers, respectively, proj_L and proj_{L-1} are the projection head mapping features to low-dimensional space, Ω_c is the pixel set of class c, and $\alpha = 0.8$ is the balancing parameter.

1) Prototype Memory Bank: We design a PMB to store feature prototypes for each class c in every image

$$\mu_{c,I} = \frac{1}{|\Omega_{c,I}|} \sum_{p \in \Omega_{c,I}} z_p \tag{12}$$

where $\Omega_{c,I}$ is the set of pixels belonging to class c in image I, and z_p is the feature representation of pixel p. The memory

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

bank size equals the sum of class counts across all training images.

At the end of each training epoch, we calculate global class prototypes based on the prototypes in the memory bank and construct a class similarity matrix

$$\mu_c = \frac{1}{|M_c|}\sum_{\mu\in M_c}\mu, \quad P_{i,j} = \frac{\mu_i \cdot \mu_j}{\|\mu_i\| \cdot \|\mu_j\|} \tag{13}$$

where M_c denotes the set of all prototypes for class c stored in the memory bank across all training images, μ_c represents the global prototype for class c, and $P_{i,j}$ is the cosine similarity between the global prototypes of classes i and j.

2) Global and Local Sampling Strategies: We design a combination of global (memory bank-based) and local (current batch-based) sampling strategies to fully utilize historical information and current feature distributions.

For an anchor point of class c, the negative sample allocation probability for other class i is calculated through the similarity matrix

$$D_c = \text{softmax}\left(P_{-c}[:,c]\right), \quad N_i^c = N_{\text{neg}} \cdot D_c[i] \tag{14}$$

where P is the class similarity matrix defined in (12), $P_{-c}[:,c]$ denotes the cth column of P excluding the diagonal element $P_{c,c}$ (representing similarities between class c and all other classes), D_c is the normalized probability distribution obtained by applying softmax to $P_{-c}[:,c]$, N_{neg} is the total number of negative samples to be selected, and N_i^c is the number of negative samples allocated to class i for anchor class c.

For local sampling negative sample quantity calculation, we compute the classes that exist in the mini-batch. During the first epoch, when the similarity matrix is not yet available, we employ a uniform allocation strategy for determining the number of negative samples across classes.

Local Sampling: Within the current batch, we employ a two-stage strategy.

1) *Candidate Sample Expansion:* For each class $i \neq c$, randomly select $2N_i^c$ candidate negative samples; for class c, randomly select $2N_p$ candidate positive samples.
2) *Similarity Filtering:* Select N_i^c negative samples with highest similarity and N_p positive samples with lowest similarity.

Memory Bank Sampling: For each anchor point z_c belonging to class c, we select from the memory bank.

1) *Positive Samples:* Select N_p same-class samples with the lowest similarity to the anchor to promote intraclass diversity learning.
2) *Negative Samples:* For each nonanchor class i, select N_i^c samples with the highest similarity to the anchor, focusing on hard negative samples.

Cosine similarity is employed to evaluate the relationship between the anchor and the samples.

3) Contrastive Loss Function: The final contrastive loss function integrates samples selected from both the memory bank and local sampling

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{|A|}\sum_{a\in A}\log\frac{\sum_{p\in\mathcal{P}(c)\cup\mathcal{P}_{\mathcal{M}}(c)}\exp\left(\text{sim}\left(z_c,z_p\right)/\tau_{\text{cl}}\right)}{Z_c}$$

TABLE I
COMPUTATIONAL COST OF PGCL (INPUT IMAGE SIZE: 512 × 512)

Component	Params (M)	FLOPs (G)	GPU Mem (MB)
Projection Heads	0.06	4.20	24
Memory Bank	4.20	0	17
Sampling & Loss	0	2.08	659
PGCL Total	4.26	6.28	700

$$\text{where } Z_c = \sum_{p\in\mathcal{P}(c)\cup\mathcal{P}_{\mathcal{M}}(c)}\exp\left(\text{sim}\left(z_c,z_p\right)/\tau_{\text{cl}}\right) + \sum_{n\in\mathcal{N}(c)\cup\mathcal{N}_{\mathcal{M}}(c)}\exp\left(\text{sim}\left(z_c,z_n\right)/\tau_{\text{cl}}\right) \tag{15}$$

where τ_{cl} is the contrastive learning temperature parameter (set to 0.07), and A is the set of anchor points in the batch.

4) Computational Cost: We provide a detailed breakdown of PGCL's resource requirements in Table I. The module adds 4.26M parameters and 6.28 giga floating point operations (GFLOPs), with the majority of parameters (4.20M) allocated to the PMB for storing class-specific embeddings. Notably, PGCL achieves sublinear memory scaling through fixed-size sampling (64 anchors regardless of batch size), making it practical for large-scale training.

D. Semantic-Change Consistency Regularization

In practical remote sensing applications, semantic change and change detection results have strict logical relationships—a location should be marked as changed only when the semantic labels from two temporal phases differ; conversely, when semantic labels remain consistent, the location should be considered as nonchanged. To address logical inconsistencies in prediction results, we design a probability-based consistency measure.

Existing multitask SCD approaches have explored logical interactions between semantic segmentation and change detection tasks through various strategies. For instance, BT-SCD [48] employs a logit-level interaction strategy that compares discrete categorical predictions from two SS branches—when $\arg\max(P_{\text{SS}}^1) = \arg\max(P_{\text{SS}}^2)$ at a position, the model uses $P_{\text{CD}}[0]$ (unchanged probability) to guide SS optimization; otherwise, it uses $P_{\text{CD}}[1]$ (changed probability), and then minimizes the mean squared error between these probabilities and the maximum SS prediction confidences. While this approach establishes explicit task connections, it relies on hard categorical decisions based on arg max operations that may not fully capture the nuanced semantic relationships in remote sensing imagery. Our method addresses this limitation by computing consistency directly from continuous semantic distributions, enabling more fine-grained modeling of semantic change patterns.

Specifically, we define the change detection probability $P_{\text{chg}}(i)$ as the probability of change occurring at position i directly predicted by the model, and the semantic change probability $P_{\text{sem}}(i) = 1 - \text{sim}(S_1(i), S_2(i))$ as the probability of semantic change at position i based on bitemporal semantic segmentation predictions, where $\text{sim}(\cdot,\cdot)$ quantifies

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

the cosine similarity between two semantic distributions. This continuous formulation provides several advantages over discrete comparison approaches. When $S_1(i)$ and $S_2(i)$ represent semantically similar classes (e.g., "tree" with confidence 0.6 and "low vegetation" with confidence 0.7), the cosine similarity $\text{sim}(S_1(i), S_2(i))$ might yield a value like 0.85, resulting in a semantic change probability $P_{\text{sem}}(i) = 0.15$ that reflects their relatedness. In contrast, the arg max-based discrete comparison would simply indicate a categorical change, losing this nuanced similarity information. This soft measure is particularly valuable in remote sensing scenarios where spectral variability and mixed pixels create ambiguous class boundaries, and where accurate detection of rare category transitions requires more sophisticated supervision signals than binary categorical comparisons can provide.

Algorithm 1 SemiSCD Training Framework

Require: Bi-temporal images (I_1, I_2) with sparse labels M
Ensure: Semantic maps S_1, S_2 and change map B

Initialize student model $\theta_{student}$, memory bank $\mathcal{M} \leftarrow \emptyset$
// Stage 1: Supervised Pre-training
for each training iteration **do**
  Extract features: $F_1, F_2 = \Phi_{encoder}(I_1, I_2)$
  Predict: $S_1, S_2 = \Phi_{decoder,SS}(F_1, F_2)$, $B = \Phi_{decoder,CD}(|F_1 - F_2|, |S_1 - S_2|)$
  // Prototype-Guided Contrastive Learning (PGCL)
  Construct anchor: $z_a = \alpha \cdot \text{proj}_L(F_L) + (1-\alpha) \cdot \text{proj}_{L-1}(\text{avg}(F_{L-1}))$
  Sample positives/negatives from batch and memory bank $\mathcal{M}$
  $\mathcal{L}_{contrast} = -\frac{1}{|A|} \sum \log \frac{\sum_p \exp(\text{sim}(z_a, z_p)/\tau)}{\sum_p \exp(\cdot) + \sum_n \exp(\cdot)}$
  // Semantic-Change Consistency Regularization (SCCR)
  $\mathcal{L}_{consist} = \frac{1}{N} \sum |P_{chg}(i) - (1 - \text{sim}(S_1[i], S_2[i]))|$
  Optimize: $\mathcal{L}_{total} = 2\mathcal{L}_{scd} + 2\mathcal{L}_{cd} + 0.2\mathcal{L}_{contrast} + 0.2\mathcal{L}_{consist}$
end for
Update memory bank: $\mu_{c,I} = \frac{1}{|\Omega_{c,I}|} \sum z_p \rightarrow \mathcal{M}$, update $P_{i,j} = \frac{\mu_i \cdot \mu_j}{\|\mu_i\| \|\mu_j\|}$
// Stage 2: Semi-supervised learning with pseudo-labels
Initialize teacher model: $\theta_{teacher} \leftarrow \theta_{student}$
for each training iteration **do**
  // Adaptive Threshold Semi-Supervised Learning (ATSSL)
  Generate pseudo-labels with teacher (no grad): Keep if confidence $> \tau_c$
  where $\tau_c = \tau_{base} \cdot \left(1 + \beta \cdot \log \frac{N_{max}}{N_c}\right)^{-1}$
  $\mathcal{L}_{ss} = \mathcal{L}_{sup} + \lambda^u_{sem} \cdot \mathcal{L}_{pseudo}$, $\lambda^u_{sem} = N_r^p / N_t^p$
  Compute $\mathcal{L}_{contrast}$ and $\mathcal{L}_{consist}$ (same as Stage 1)
  Optimize: $\mathcal{L}_{total} = 2\mathcal{L}_{scd} + 2\mathcal{L}_{cd} + \mathcal{L}_{ss} + 0.2\mathcal{L}_{contrast} + 0.2\mathcal{L}_{consist}$
  Update teacher: $\theta_{teacher} \leftarrow \alpha \cdot \theta_{teacher} + (1-\alpha) \cdot \theta_{student}$
end for
Update $\mathcal{M}$, $P_{i,j}$ and adapt τ_{base} based on validation accuracy
return S_1, S_2, B

Based on these two probabilities, we define the inconsistency measure $\Delta S(i) = P_{\text{chg}}(i) - P_{\text{sem}}(i)$, which should ideally be close to zero. The semantic-change consistency loss is defined as the absolute mean of the inconsistency measure: $\mathcal{L}_{\text{consist}} = 1/N \sum_{i=1}^{N} |\Delta S(i)|$. This regularization term ensures that change detection predictions maintain logical consistency with semantic change predictions, thereby improving the reliability of model prediction results.

E. Total Loss Function

The total loss function is formulated as a weighted combination

$$\mathcal{L}_{\text{total}} = 2\mathcal{L}_{\text{scd}} + 2\mathcal{L}_{\text{cd}} + \mathcal{L}_{\text{ss}} + 0.2\mathcal{L}_{\text{contrast}} + 0.2\mathcal{L}_{\text{consist}}. \quad (16)$$

Here, $\mathcal{L}_{\text{scd}}$ and $\mathcal{L}_{\text{cd}}$ represent the cross-entropy losses for the semantic segmentation and change detection branches, respectively.

F. Overall Training Procedure

To provide a comprehensive view of our framework, Algorithm 1 summarizes the complete training procedure of SemiSCD, integrating all the proposed components including ATSSL, PGCL, and SCCR.

IV. Experiments

A. Datasets

The SECOND dataset consists of 4662 pairs of bitemporal remote sensing images, each with dimensions of 512×512 pixels and a spatial resolution of 0.5 m. The dataset contains the following semantic categories: no change, low vegetation, building, tree, water, playground, and ground, totaling seven categories.

The Landsat-SCD dataset is based on multitemporal data collected by the Landsat satellite series, with a spatial resolution of 30 m. This dataset contains 8468 pairs of images with dimensions of 416×416 pixels, primarily distributed across agricultural and urban areas in multiple provinces of China. The dataset includes one "no change" category and four types of land cover: farmland, desert, building, and water. The dataset is divided into training, validation, and testing sets in the ratio of 8:1:1.

The HiU-CD mini dataset [49] focuses on urban SCD with ultrahigh spatial resolution. This dataset uses aerial images with a spatial resolution of 0.1 m captured in Tallinn, Estonia, spanning three time phases (2017, 2018, and 2019) and consists of 745 image pairs with dimensions of 1024×1024 pixels (cropped to 512×512 for training). The dataset comprises nine semantic classes: water, grass, building, greenhouse, road, bridge, bare land, woodland, and others.

B. Evaluation Metrics

To comprehensively assess our method's performance, we adopt four complementary evaluation metrics that measure different aspects of model effectiveness:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

1) mIoU [50]: The mean intersection over union (mIoU) is the average of the IoU values between the invariant region ($\mathrm{IoU_{nc}}$) and the changing region (IoU_c)

$$\mathrm{mIoU} = \frac{\mathrm{IoU_{nc}} + \mathrm{IoU}_c}{2} \tag{17}$$

$$\mathrm{IoU_{nc}} = \frac{q_{00}}{\sum_{i=0}^{N} q_{i0} + \sum_{j=0}^{N} q_{0j} - q_{00}} \tag{18}$$

$$\mathrm{IoU}_c = \frac{\sum_{i=1}^{N}\sum_{j=1}^{N} q_{ij}}{\left(\sum_{i=0}^{N}\sum_{j=0}^{N} q_{ij} - q_{00}\right)} \tag{19}$$

where q_{ij} represents the element in the confusion matrix at row i and column j, index 0 represents the unchanged category, and indices 1–N represent change categories.

2) SeK [51]: This metric specifically assesses the model's ability to recognize rare categories, considering class imbalance in classification performance. It is calculated as

$$\mathrm{SeK} = \mathrm{eIoU}_{c-1} \cdot \frac{\rho - \eta}{1 - \eta} \tag{20}$$

where $\hat{q}_{ij}$ represents the element in the confusion matrix at row i and column j, $\rho = \left(\sum_i \hat{q}_{ii}\right) / \left(\sum_{i,j} \hat{q}_{ij}\right)$ represents the overall accuracy (OA, the sum of diagonal elements divided by the sum of all elements in the confusion matrix), $\eta = \sum_i \left(\sum_j \hat{q}_{ij} \sum_j \hat{q}_{\mathrm{ji}}\right) / \left(\left(\sum_{i,j} \hat{q}_{ij}\right)^2\right)$ represents the consistency produced by class frequency imbalance (accounting for the expected agreement by chance), and eIoU_{c-1} represents the mean IoU of the $c-1$ most rare categories.

3) F_{SCD} [21]: This metric evaluates the comprehensive performance of change detection, focusing on the classification accuracy of change regions. F_{SCD} is calculated by computing the precision (P_{SCD}) and recall (R_{SCD}) of change regions, and then calculating their harmonic mean

$$P_{\mathrm{SCD}} = \frac{\sum_{i=1}^{N} q_{ii}}{\sum_{i=1}^{N}\sum_{j=0}^{N} q_{ij}} \tag{21}$$

$$R_{\mathrm{SCD}} = \frac{\sum_{i=1}^{N} q_{ii}}{\sum_{i=0}^{N}\sum_{j=1}^{N} q_{ij}} \tag{22}$$

$$F_{\mathrm{SCD}} = \frac{2 \times P_{\mathrm{SCD}} \times R_{\mathrm{SCD}}}{P_{\mathrm{SCD}} + R_{\mathrm{SCD}}} \tag{23}$$

where P_{SCD} represents the proportion of true change regions among all regions predicted as changed, while R_{SCD} represents the proportion of correctly detected change regions among all actual change regions.

4) OA [52]: OA measures the overall pixelwise classification accuracy across all categories, providing a holistic assessment of model performance. It is particularly valuable for evaluating performance consistency across datasets with different class distributions. OA is calculated as

$$\mathrm{OA} = \frac{\sum_{i=0}^{N} q_{ii}}{\sum_{i=0}^{N}\sum_{j=0}^{N} q_{ij}} \tag{24}$$

where $\sum_{i=0}^{N} q_{ii}$ represents the total number of correctly classified pixels, and $\sum_{i=0}^{N}\sum_{j=0}^{N} q_{ij}$ represents the total number of pixels.

C. Implementation Details

All experiments are conducted on servers equipped with NVIDIA Tesla A100 GPUs. The batch size is set to 8. For the SECOND dataset and HiU-CD mini dataset, we train the model for a total of 80 epochs: the first 45 epochs employ fully supervised learning using only labeled data to establish stable feature representations, after which semi-supervised learning is introduced from epoch 45 to leverage unlabeled data. For the Landsat-SCD dataset, training extends to 120 epochs total, with fully supervised learning for the first 80 epochs and semi-supervised learning from epoch 80 onward. We employ the AdamW optimizer with a weight decay coefficient of 0.0001 to enhance model generalization ability. The learning rate schedule follows a poly strategy with an initial learning rate of $1e^{-4}$ for the backbone network. The learning rate dynamically decays according to the formula $\mathrm{lr} = \mathrm{baselr} \times (1 - \mathrm{iter}/\mathrm{totaliter})^{0.9}$, ensuring an effective and stable training process. For the contrastive learning module, 64 anchor points are randomly selected in each mini-batch for feature learning. To enhance model robustness and prevent overfitting, we implement multiple data augmentation strategies, including horizontal flipping, vertical flipping, random rotation, and random color jittering.

D. Comparison With Sota

Table II presents our comparative experimental results, highlighting SemiSCD's superior performance across evaluation metrics on three diverse datasets.

On the SECOND dataset, our method achieves an mIoU of 74.36%, SeK of 26.38%, F_{SCD} of 63.89%, and OA of 89.67%, surpassing the previous best method SSCLNet by 1.47%, 0.87%, 0.47%, and 1.93%, respectively. The SeK improvement demonstrates our method's enhanced capability in recognizing rare categories such as water and playground.

On the Landsat-SCD dataset, which features longer temporal spans and more complex change patterns, the advantages of SemiSCD become even more pronounced. Our method achieves 91.74% mIoU, 66.84% SeK, 89.13% F_{SCD}, and 98.74% OA, with improvements of 1.03%, 1.95%, 2.95%, and 0.73% over SSCLNet, respectively.

On the HiU-CD mini dataset, which presents additional challenges with dense urban scenes and fine-grained semantic categories, SemiSCD achieves 70.44% mIoU, 18.27% SeK, 56.23% F_{SCD}, and 92.11% OA, outperforming SSCLNet by 1.23%, 0.61%, 0.86%, and 0.75%, respectively. The consistent improvements across all three datasets—spanning different spatial resolutions, scene types, and complexity levels—validate the generalizability and robustness of our approach. Notably, our method achieves SeK improvements across all datasets (0.87% on SECOND, 1.95% on Landsat-SCD, and 0.61% on HiU-CD mini), demonstrating that the combination of PGCL and adaptive threshold mechanisms effectively addresses rare category recognition challenges across diverse data distributions. Meanwhile, the consistent OA leadership across all three datasets (89.67%, 98.74%, and 92.11%) indicates that our framework enhances overall

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTITATIVE COMPARISON WITH OTHER METHODS

Method	SECOND				Landsat-SCD				HiU-CD mini			
	mIoU↑	SeK↑	F_{SCD} ↑	OA↑	mIoU↑	SeK↑	F_{SCD} ↑	OA↑	mIoU↑	SeK↑	F_{SCD} ↑	OA↑
SMNet [53]	71.15	20.29	60.42	86.73	85.73	51.22	85.55	94.46	63.24	9.55	42.37	85.33
HRSCD-str.4 [51]	71.07	18.72	58.14	85.99	79.02	32.21	73.26	97.23	62.23	6.44	37.32	83.12
MTSCDNet [15]	71.77	20.65	58.34	86.21	89.55	64.78	88.41	97.02	67.73	15.24	53.27	89.91
ChangeMask [50]	71.35	17.81	57.38	86.46	83.58	47.25	79.39	91.79	67.32	17.32	55.24	88.45
SCDNet [54]	73.14	23.74	60.09	88.21	85.54	63.35	86.40	93.87	65.32	13.21	48.23	85.22
Bi-SRNet [1]	73.33	23.14	62.53	88.45	85.18	50.93	84.96	94.01	64.71	10.26	46.24	84.22
SCanNet [55]	73.29	23.82	63.54	88.23	88.85	60.45	89.17	96.15	66.45	15.84	52.66	89.01
MambaSCD [26]	73.59	22.84	62.76	87.33	86.64	53.08	65.34	95.34	68.46	17.23	54.66	91.01
DEFO-MTSCD [25]	73.62	23.61	62.62	88.46	85.76	48.54	86.31	94.25	67.65	16.84	54.36	90.86
SSCLNet [56]	72.89	25.51	63.42	87.74	90.71	64.89	86.18	98.01	69.21	17.66	55.37	91.36
SemiSCD (Ours)	**74.36**	**26.38**	**63.89**	**89.67**	**91.74**	**66.84**	**89.13**	**98.74**	**70.44**	**18.27**	**56.23**	**92.11**

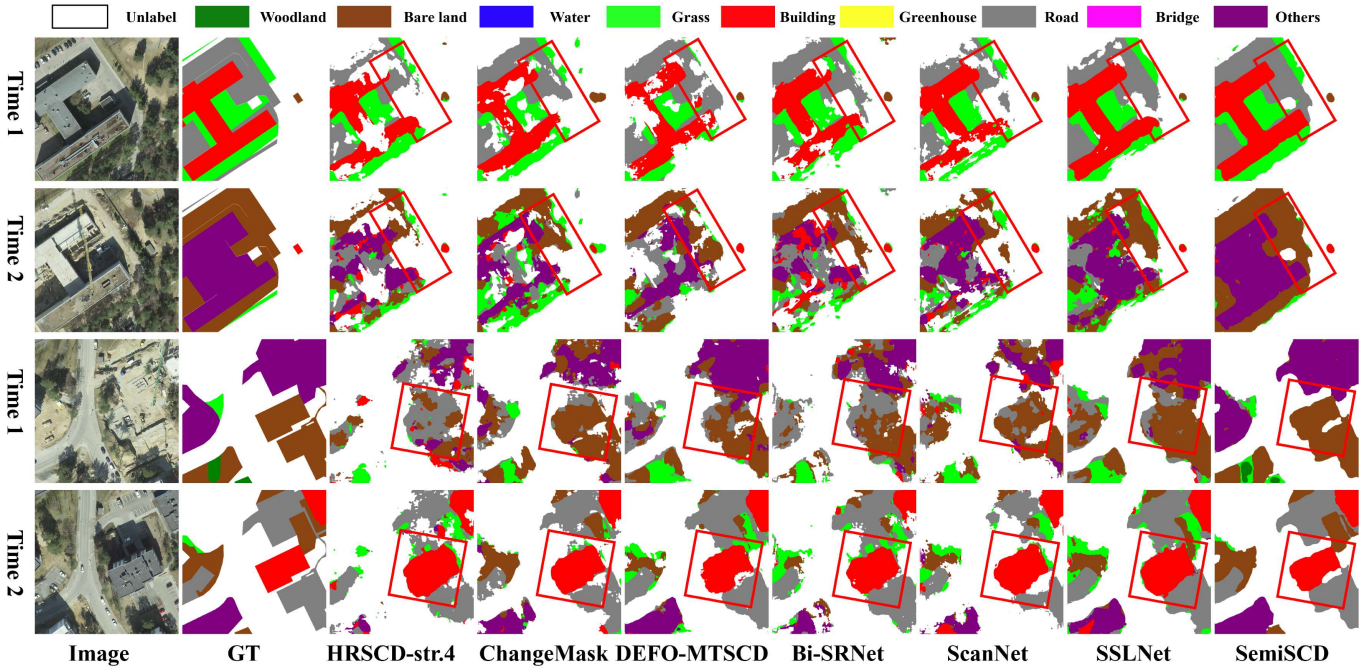


Fig. 5. Qualitative comparison of SCD results on the HiU-CD mini dataset.

prediction quality while maintaining strong performance on minority classes.

Fig. 5 presents qualitative results on the HiU-CD mini dataset, where SemiSCD achieves more accurate semantic segmentation and cleaner change boundaries compared to baseline methods. As indicated by the red boxes in rows 1–2 and 3–4, our method demonstrates superior performance in distinguishing visually similar categories (e.g., bare land versus road) and handling complex building structures with shadows and occlusions. Fig. 6 visualizes the comparative experimental results on the SECOND dataset. In the third and fourth rows of Fig. 6, most existing methods exhibit inconsistencies between semantic segmentation and change detection results. For instance, ChangeMask and DEFO-MTSCD show semantic segmentation results indicating change in certain regions, while the change map indicates no change, or vice versa. SemiSCD effectively addresses this issue through our SCCR mechanism. Fig. 7 presents the comparative experimental results on the Landsat-SCD dataset. As indicated by the red boxes in rows 1–2 and 3–4, SemiSCD demonstrates superior performance in distinguishing spectrally similar land cover transitions and complex change patterns compared to other methods.

Fig. 8 presents confusion matrix visualizations for three representative methods on the SECOND and Landsat-SCD datasets. On the SECOND dataset (top row), SemiSCD demonstrates superior diagonal values across all categories, particularly for rare classes such as water and playground. The water class achieves 0.548 accuracy with SemiSCD, compared to 0.498 for SCanNet and 0.527 for SSCLNet, while playground reaches 0.606 with our method versus 0.548 for SCanNet and 0.581 for SSCLNet. The confusion matrices reveal that existing methods frequently misclassify tree and low vegetation due to their visual similarity, whereas SemiSCD achieves better discrimination with reduced off-diagonal confusion. On the Landsat-SCD dataset (bottom row), SemiSCD

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

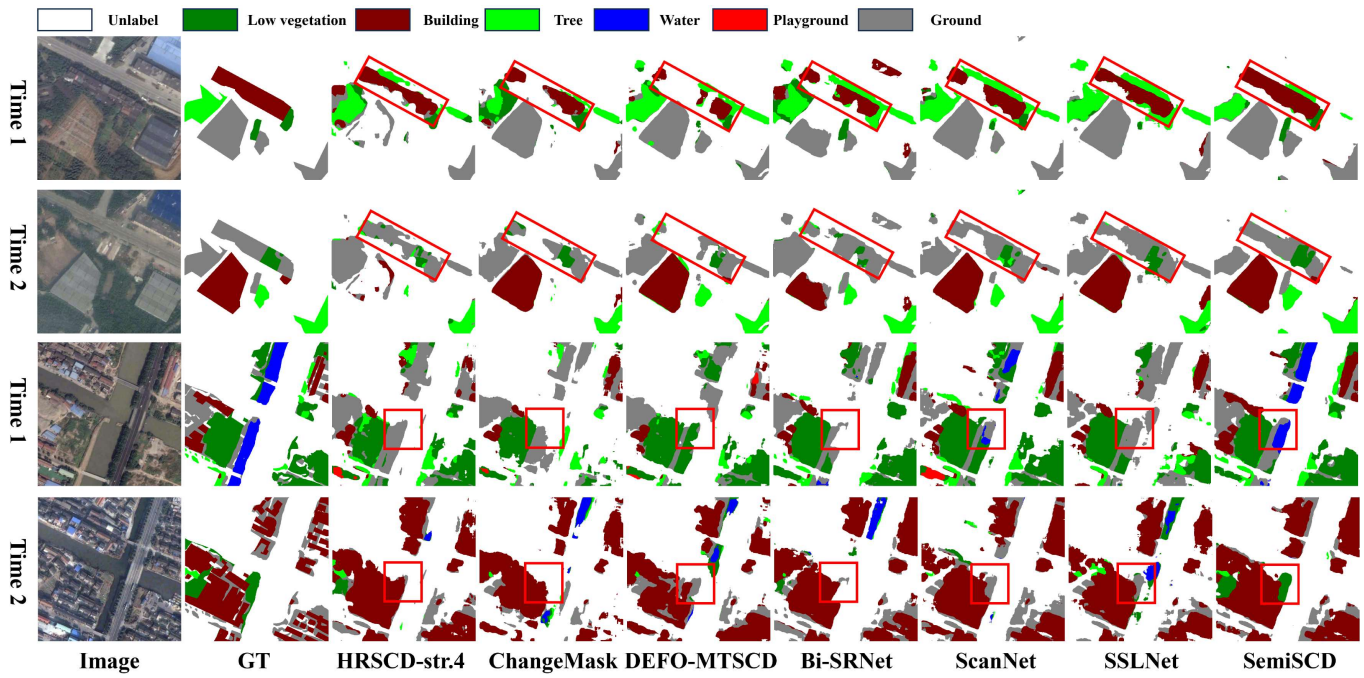


Fig. 6. Qualitative comparison of SCD results on the SECOND dataset.

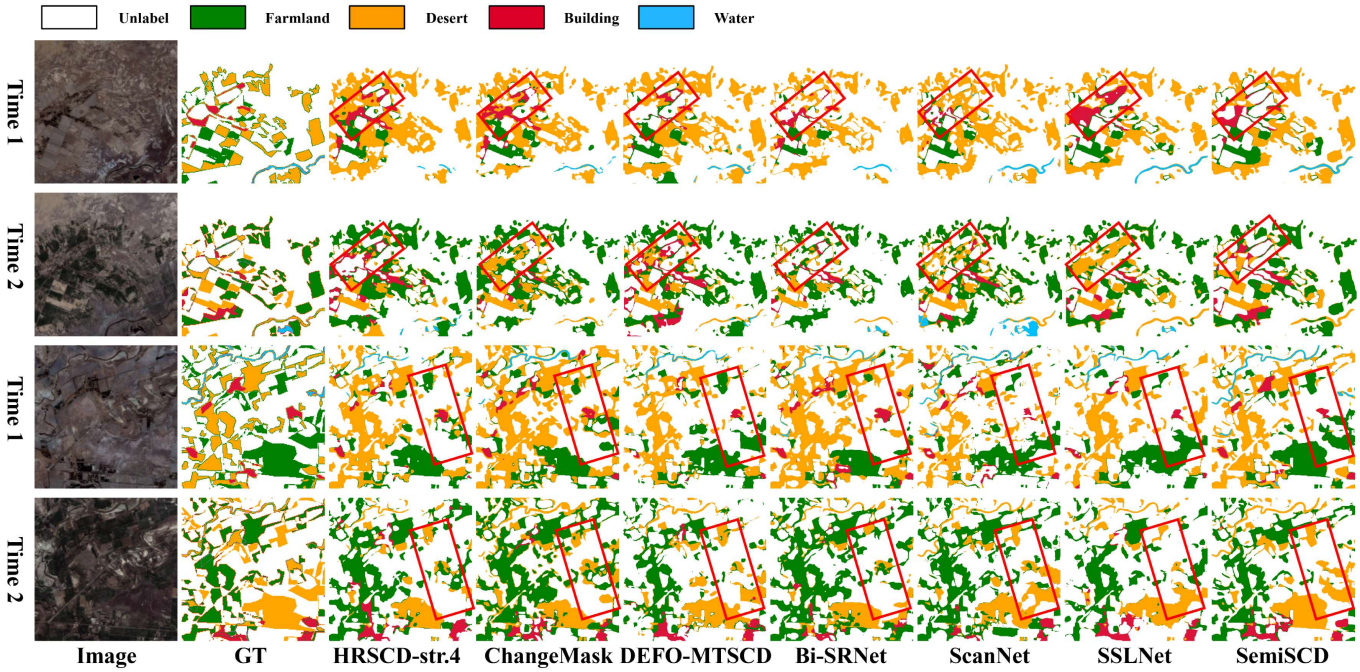


Fig. 7. Qualitative comparison of SCD results on the Landsat-SCD dataset.

consistently outperforms competitors across all categories, achieving 0.985 for no change, 0.899 for farmland, 0.918 for desert, 0.912 for building, and 0.950 for water. The most notable improvement appears in farmland classification, where SemiSCD exhibits a misclassification rate of only 0.082 to no change, substantially lower than SCanNet at 0.135 and SSCLNet at 0.120.

E. Ablation Studies

1) Analysis of Main Components: As shown in Table III, on the SECOND dataset, experiments based on the HRNet40 backbone demonstrate that the complete SemiSCD framework achieves 74.36% mIoU, 26.38% SeK, and 63.89% F_{SCD}. Notably, with the introduction of the SCCR component, model performance exhibits initial enhancement; when further integrated with the ATSSL component, mIoU increases to 73.15% and SeK reaches 25.08%; finally, after incorporating the PGCL component, model performance attains optimality. Analysis of class-specific $F1$ scores in Table IV on the SECOND dataset reveals that the SemiSCD framework achieves consistent performance improvements across all categories. Particularly noteworthy is that for rare categories with extremely low proportions, such as water (0.26%) and playground (0.12%), the complete SemiSCD method improves $F1$ scores by 3.20%

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

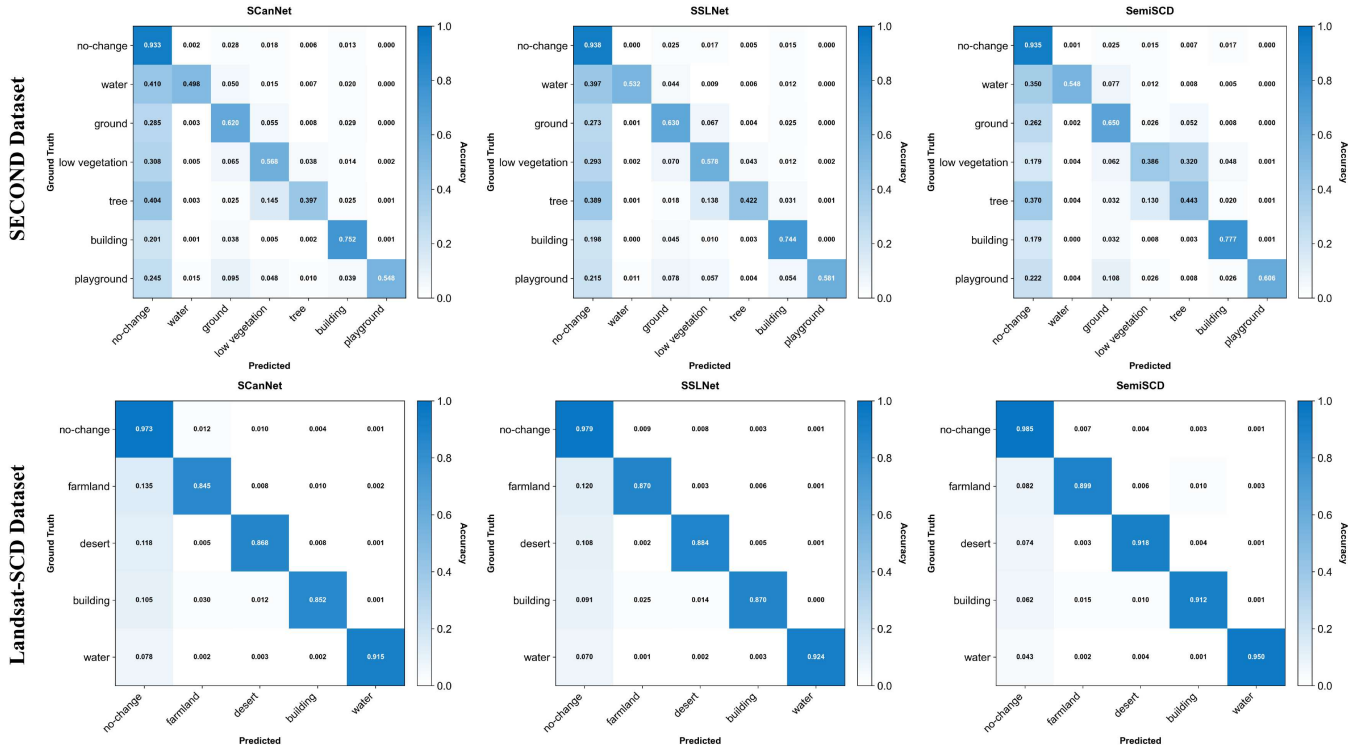


Fig. 8. Confusion matrix comparison on SECOND and Landsat-SCD datasets.

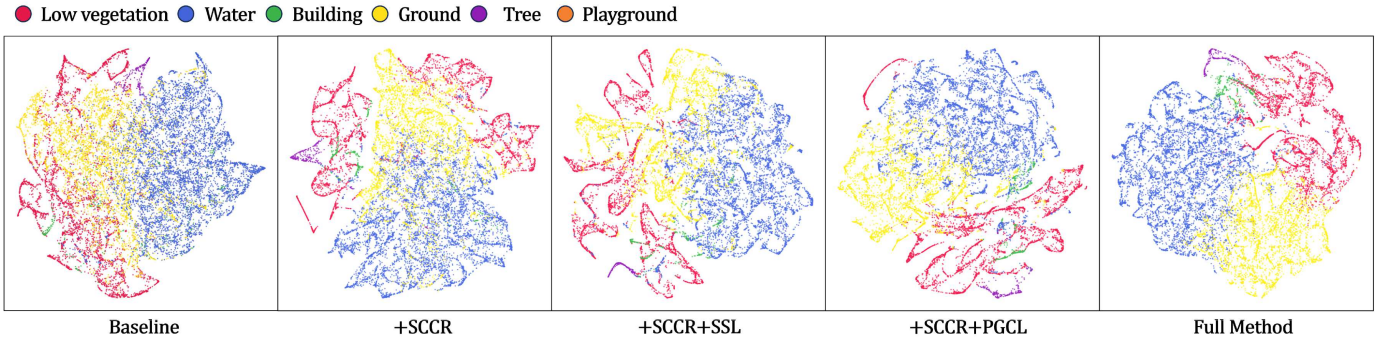


Fig. 9. t-SNE visualization of feature distributions with different model combinations on the SECOND dataset.

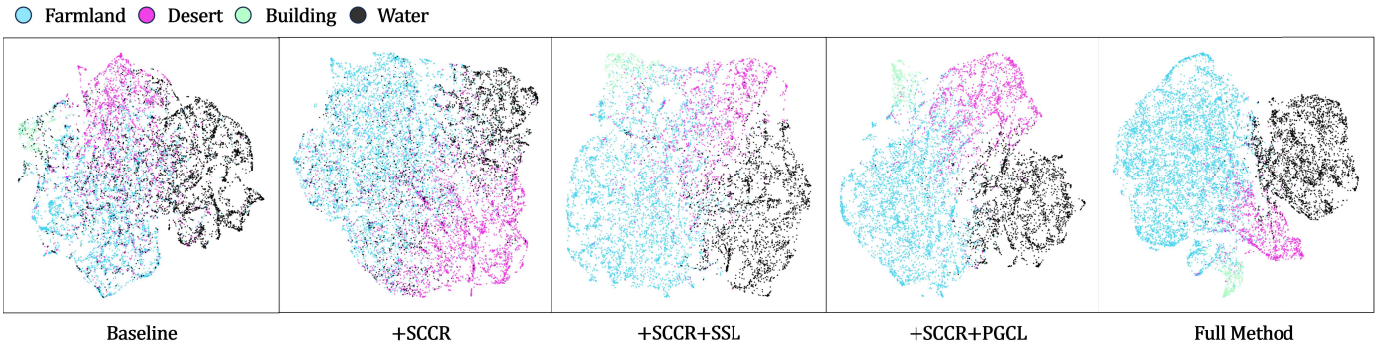


Fig. 10. t-SNE visualization of feature distributions with different model combinations on Landsat-SCD dataset.

and 3.49% compared to the baseline, respectively. Figs. 9 and 10 present the t-distributed stochastic neighbor embedding (t-SNE) visualization of feature distributions, clearly illustrating that as components are progressively added, the feature dis-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ANALYSIS OF THE MAIN ABLATION STUDIES (%)

Method	SECOND			Landsat-SCD		
	mIoU↑	SeK↑	F_{SCD} ↑	mIoU↑	SeK↑	F_{SCD} ↑
Baseline	72.15	23.52	60.35	88.02	63.08	86.17
+ SCCR	72.38	23.79	61.37	88.76	64.02	86.61
+ SCCR + ATSSL	73.15	25.08	62.69	89.95	65.90	88.25
+ SCCR + PGCL	73.71	25.60	63.13	90.46	66.07	88.57
Full Method	**74.36**	**26.38**	**63.89**	**91.74**	**66.84**	**89.13**

TABLE IV
ABLATION STUDY ON CLASS-SPECIFIC $F1$ SCORES. THE PERCENTAGE INDICATES THE PROPORTION OF EACH CLASS IN THE DATASET (%)

Method	F1					
	Ground (7.18%)	Building (5.88%)	Low veg. (4.94%)	Tree (1.56%)	Water (0.26%)	Playground (0.12%)
Baseline	62.51	74.33	60.08	49.44	51.69	51.51
+ SCCR + ATSSL	64.09	75.22	60.99	50.32	54.07	53.96
+ SCCR + PGCL	64.31	75.44	60.71	50.99	53.59	53.07
Full Method	**65.08**	**76.31**	**61.85**	**51.46**	**54.89**	**55.00**

TABLE V
ABLATION STUDY ON ATSSL. COMPARISON OF ADAPTIVE THRESHOLD STRATEGY WITH FIXED THRESHOLD APPROACHES (%)

Method	SECOND			Landsat-SCD		
	mIoU↑	SeK↑	F_{SCD} ↑	mIoU↑	SeK↑	F_{SCD} ↑
Baseline	72.38	23.79	61.37	88.76	64.02	86.61
+SSL(Fixed 0.4)	72.45	24.37	62.35	88.94	64.23	86.81
+SSL(Fixed 0.5)	72.79	24.55	62.33	89.23	64.57	87.23
+SSL(Fixed 0.6)	73.01	24.93	62.48	89.79	65.02	87.87
+SSL(Fixed 0.7)	72.50	24.81	62.57	89.36	65.21	87.54
+ATSSL	**73.15**	**25.08**	**62.69**	**89.95**	**65.90**	**88.25**

TABLE VI
ABLATION STUDY ON DIFFERENT COMPONENTS OF THE PGCL. FA: FEATURE FUSION ANCHOR; PSG: PROTOTYPE SIMILARITY-GUIDED SAMPLING; LC: LOCAL CONTRASTIVE LEARNING; AND GC: GLOBAL CONTRASTIVE LEARNING (%)

FA	PSG	LC	GC	mIoU↑	SeK↑	F_{SCD} ↑
✓	✓	✓	✓	**73.71**	**25.60**	**63.13**
✓	×	✓	✓	73.54	25.52	63.01
✓	✓	×	✓	73.15	24.94	62.53
✓	✓	✓	×	73.37	25.09	62.57
×	✓	✓	✓	73.46	25.31	62.84
×	×	×	×	72.38	23.79	61.37

tributions form more compact intraclass clusters and exhibit clearer interclass separations.

2) Analysis of ATSSL: Table V presents the comparison results between the adaptive threshold strategy and fixed threshold approaches on both the SECOND and Landsat-SCD datasets. While the semi-supervised learning method with fixed thresholds achieves significant performance improvements compared to the baseline methods, the introduction of ATSSL further enhances the performance.

3) Analysis of PGCL: To evaluate the effectiveness of the PGCL framework, we conducted ablation experiments as shown in Table VI. The complete PGCL achieves 73.71% mIoU, 25.60% SeK, and 63.13% F_{SCD} on the SECOND dataset, representing improvements of 1.33%, 1.81%, and 1.76% over the baseline model, respectively. The ablation experiments demonstrate the effectiveness of all components: removing GC, LC, PSG, and FA leads to varying degrees of performance degradation, highlighting the necessity and contribution of each component. As demonstrated in Table VII, we conducted a comprehensive analysis of the feature fusion anchor parameter α to determine its optimal value. The results show that $\alpha = 0.8$ yields the best performance with 73.71% mIoU. This finding indicates that prioritizing final layer semantic information (80%) while incorporating complementary intermediate spatial details (20%) establishes an effective balance between high-level semantic understanding and precise spatial localization.

TABLE VII
ABLATION STUDY ON FA PARAMETER α (%)

α	0.2	0.4	0.6	**0.8**	1.0
mIoU↑	72.77	73.09	73.48	**73.71**	73.38

TABLE VIII
COMPUTATIONAL EFFICIENCY COMPARISON ON SECOND DATASET. T-TIMES DENOTES THE TRAINING TIME, AND I-TIMES DENOTES THE INFERENCE TIME

Method	Params (M)↓	FLOPs (G)↓	T-Times (S)↓	I-Times (ms)↓	SeK ↑
BI-SRNet	23.38	190.30	356	79	23.14
MambaSCD	54.28	146.70	577	111	22.84
SCDNet	39.62	116.98	320	87	23.74
MTSCD	94.62	290.28	754	175	20.65
SSCLNet	48.20	122.24	410	94	25.51
SemiSCD (Ours)	51.78	133.45	438	98	26.38

F. Efficiency Analysis

Table VIII presents the computational efficiency comparison on the SECOND dataset under our experimental settings. SemiSCD achieves 74.36% mIoU and 26.38% SeK with 51.78M parameters and 133.45 GFLOPs, requiring 438 s/epoch for training and 98 ms for inference. While our framework introduces an additional 3.58M parameters compared to SSCLNet (48.20M), this modest increase is well-justified by the substantial performance gains, particularly in rare category recognition, where SemiSCD achieves 26.38% SeK versus 25.51% for SSCLNet.

G. Ablation Studies on ResNet34 Backbone

To validate the generalization capability of our proposed method, we conducted comparative experiments using ResNet34 as the backbone network. As shown in Table IX, the complete SemiSCD framework achieves consistent improvements across all metrics on both datasets. On the SECOND dataset, our full method achieves 72.51% mIoU, 24.58% SeK, and 62.37% F_{SCD}, demonstrating improvements of 2.22%, 2.59%, and 3.33% compared to the baseline, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
ABLATION STUDIES ON RESNET34 BACKBONE (%)

Method	SECOND			Landsat-SCD		
	mIoU↑	SeK↑	F_{SCD} ↑	mIoU↑	SeK↑	F_{SCD} ↑
ResNet34						
Baseline	70.29	21.99	59.04	85.43	60.75	84.21
+ SCCR	70.57	22.20	59.88	86.18	61.64	84.93
+ SCCR + ATSSL	71.39	23.24	60.96	87.65	63.54	86.37
+ SCCR + PGCL	71.82	23.87	61.46	88.12	63.86	86.92
Full Method	**72.51**	**24.58**	**62.37**	**89.31**	**64.53**	**87.84**

TABLE X
ABLATION STUDY ON SEMI-SUPERVISED LEARNING START EPOCH (%)

Start Epoch	35	40	45	50	55
mIoU↑	73.92	74.08	**74.36**	74.30	74.16
SeK↑	25.26	25.45	**26.38**	26.21	26.03
F_{SCD} ↑	63.59	63.72	**63.89**	63.85	63.78

Similarly, on the Landsat-SCD dataset, the full method attains 89.31% mIoU, 64.53% SeK, and 87.84% F_{SCD}, showing improvements of 3.88%, 3.78%, and 3.63% over the baseline. These results confirm that our proposed components (SCCR, ATSSL, and PGCL) provide consistent benefits across different backbone architectures, validating the robustness and generalization capability of the SemiSCD framework.

H. Semi-Supervised Learning Start Epoch Analysis

As demonstrated in Table X, we investigated the optimal timing for introducing semi-supervised learning. The results show that starting at epoch 45 achieves the best performance with 74.36% mIoU, 26.38% SeK, and 63.89% F_{SCD}. Starting too early (epochs 35–40) leads to suboptimal performance due to immature feature representations, with mIoU of 73.92% and 74.08%, respectively, while starting too late (epochs 50–55) provides insufficient time for the model to benefit from unlabeled data, resulting in slightly decreased performance with mIoU of 74.30% and 74.16%. Epoch 45 represents an optimal balance where the model has established stable feature representations through supervised learning while maintaining sufficient plasticity to leverage pseudo-labels effectively.

I. Loss Weight Analysis

To investigate the impact of different loss term weights on model performance, we conducted comprehensive ablation experiments on both SECOND and Landsat-SCD datasets, as shown in Table XI. Our final configuration assigns equal weights ($\lambda_{\text{scd}} = \lambda_{\text{cd}} = 2$) to the semantic segmentation and change detection tasks, reflecting their equal importance in the SCD problem.

1) Main Task Weights (λ_{scd} and λ_{cd}): The results demonstrate that balanced weighting between the two main tasks is crucial across both datasets. When reducing all weights to 1 (uniform weighting), the model achieves only 73.25% mIoU on SECOND and 90.18% on Landsat-SCD, indicating that the main tasks require stronger supervision signals. Conversely, over-emphasizing the main tasks with $\lambda_{\text{scd}} = \lambda_{\text{cd}} = 3$ leads to 73.82% mIoU on SECOND and 91.05% on Landsat-SCD, showing marginal improvements but suggesting diminishing returns and potential overfitting risks. Our choice of $\lambda_{\text{scd}} = \lambda_{\text{cd}} = 2$ achieves the best balance, yielding 74.36% mIoU on SECOND (1.11% improvement over uniform weighting) and 91.74% on Landsat-SCD (1.56% improvement).

2) Semi-Supervised Learning Weight (λ_{ss}): The semi-supervised learning component weight significantly impacts model performance. Reducing λ_{ss} from 1 to 0.5 results in performance drops to 73.68% mIoU on SECOND and 90.85% on Landsat-SCD, indicating insufficient utilization of pseudo-label information. Conversely, increasing λ_{ss} to 1.5 yields 73.47% and 90.52% respectively, suggesting that over-reliance on pseudo-labels introduces noise and degrades performance. The optimal value $\lambda_{\text{ss}} = 1$ strikes an effective balance between leveraging unlabeled data and maintaining supervised learning stability.

3) Contrastive and Consistency Weights ($\lambda_{\text{contrast}}$ and λ_{consist}): For the auxiliary losses, we find that $\lambda_{\text{contrast}} = \lambda_{\text{consist}} = 0.2$ provides optimal performance across both datasets. Reducing these weights to 0.1 results in 73.91% mIoU on SECOND and 91.23% on Landsat-SCD, indicating insufficient regularization from the PGCL and semantic-change consistency constraints. Increasing to 0.5 yields 73.54% and 90.67%, respectively, suggesting that excessive regularization constrains the model's learning capacity and limits its ability to fit the training data effectively. Further exploration with $\lambda_{\text{contrast}} = \lambda_{\text{consist}} = 0.3$ achieves intermediate performance (73.69% and 90.89%), confirming that 0.2 represents the optimal regularization strength.

These comprehensive findings validate our loss weighting strategy and demonstrate that the proposed configuration $\{\lambda_{\text{scd}} = 2, \lambda_{\text{cd}} = 2, \lambda_{\text{ss}} = 1, \lambda_{\text{contrast}} = 0.2, \lambda_{\text{consist}} = 0.2\}$ achieves the best tradeoff among all objectives across diverse remote sensing scenarios, enabling the model to effectively learn from both labeled and unlabeled data while maintaining semantic-change consistency.

J. Robustness Analysis

To evaluate the robustness of our method, we conduct experiments by adding Gaussian noise ($\mu = 0$ and $\sigma = 0.05$) to normalized test images on the SECOND dataset. Table XII presents the performance comparison under clean and noisy conditions. The results demonstrate that SemiSCD exhibits superior robustness compared to baseline methods. Under noise perturbation, our method achieves the smallest performance degradation with only 2.31% mIoU drop, significantly outperforming SCanNet (3.18% drop) and SSCLNet (3.54% drop). This represents a 27% and 35% reduction in performance degradation compared to the two baselines, respectively. Similar trends are observed in the SeK metric, where SemiSCD maintains 2.89% degradation versus 3.62% and 4.12% for the baselines. The superior robustness can be attributed to our PGCL module, which constructs a more stable feature space by clustering semantically similar features through prototype guidance, making learned representations less sensitive to pixel-level noise perturbations. Additionally,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
ABLATION STUDY ON LOSS WEIGHT CONFIGURATIONS (%)

λ_{scd}	λ_{cd}	λ_{ss}	$\lambda_{contrast}$	$\lambda_{consist}$	SECOND			Landsat-SCD		
					mIoU↑	SeK↑	F_{SCD} ↑	mIoU↑	SeK↑	F_{SCD} ↑
1	1	1	0.2	0.2	73.25	24.87	62.34	90.18	64.92	87.45
2	2	0.5	0.2	0.2	73.68	25.42	62.89	90.85	65.47	88.26
2	**2**	**1**	**0.2**	**0.2**	**74.36**	**26.38**	**63.89**	**91.74**	**66.84**	**89.13**
2	2	1	0.1	0.1	73.91	25.76	63.12	91.23	66.15	88.54
2	2	1	0.5	0.5	73.54	25.38	62.67	90.67	65.28	87.89
3	3	1	0.2	0.2	73.82	25.61	63.28	91.05	65.89	88.37
2	2	1.5	0.2	0.2	73.47	25.29	62.58	90.52	65.13	87.76
2	2	1	0.3	0.3	73.69	25.48	62.95	90.89	65.52	88.18

TABLE XII
ROBUSTNESS COMPARISON ON SECOND DATASET WITH GAUSSIAN NOISE (σ = 0.05)

Method	Setting	mIoU (%)	SeK (%)	F_{SCD} (%)	OA (%)
SCanNet	Original	73.29	23.82	63.54	88.23
	w/ Noise	70.11 **(-3.18)**	20.20 **(-3.62)**	60.38 **(-3.16)**	85.69 **(-2.54)**
SSCLNet	Original	72.89	25.51	63.42	87.74
	w/ Noise	69.35 **(-3.54)**	21.39 **(-4.12)**	59.87 **(-3.55)**	84.83 **(-2.91)**
SemiSCD	Original	**74.36**	**26.38**	**63.89**	**89.67**
	w/ Noise	**72.05 (-2.31)**	**23.49 (-2.89)**	**61.24 (-2.65)**	**87.80 (-1.87)**

the semi-supervised learning framework enhances generalization by leveraging diverse feature distributions from both labeled and unlabeled data, improving the model's resilience to noisy inputs commonly encountered in real-world remote sensing applications.

V. CONCLUSION

In this article, we propose the SemiSCD framework, a novel approach that innovatively integrates semi-supervised learning with prototype contrastive learning for SCD in remote sensing images. We address three key challenges in remote sensing SCD: scarcity of annotated data, extreme class distribution imbalance, and complex change pattern recognition. By effectively combining adaptive threshold-guided semi-supervised learning, prototype similarity-driven feature representation optimization, and semantic-change consistency constraints, the SemiSCD framework successfully reshapes the feature space distribution structure and enhances the model's discriminative capability for rare categories and easily confused classes, while ensuring logical consistency in prediction results. Extensive experiments on two public benchmark datasets, SECOND and Landsat-SCD, demonstrate that the proposed SemiSCD framework consistently outperforms existing state-of-the-art methods across all evaluation metrics, notably exhibiting significant advantages in complex change scenarios and under extremely imbalanced class distributions. Future work will further explore temporal consistency constraints and cross-dataset domain adaptation capabilities to enhance model generalization performance in broader practical application scenarios.

REFERENCES

[1] L. Ding, H. Guo, S. Liu, L. Mou, J. Zhang, and L. Bruzzone, "Bi-temporal semantic reasoning for the semantic change detection in HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620014.

[2] J. Cheng, C. Deng, Y. Su, Z. An, and Q. Wang, "Methods and datasets on semantic segmentation for unmanned aerial vehicle remote sensing images: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 1–34, May 2024.

[3] Q. Zhu, X. Guo, Z. Li, and D. Li, "A review of multi-class change detection for satellite remote sensing imagery," *Geo-Spatial Inf. Sci.*, vol. 27, no. 1, pp. 1–15, Jan. 2024.

[4] H. He, J. Yan, D. Liang, Z. Sun, J. Li, and L. Wang, "Time-series land cover change detection using deep learning-based temporal semantic segmentation," *Remote Sens. Environ.*, vol. 305, May 2024, Art. no. 114101.

[5] Y. Himeur, B. Rimal, A. Tiwary, and A. Amira, "Using artificial intelligence and data fusion for environmental monitoring: A review and future perspectives," *Inf. Fusion*, vols. 86–87, pp. 44–75, Oct. 2022.

[6] S. Tian, Y. Zhong, Z. Zheng, A. Ma, X. Tan, and L. Zhang, "Large-scale deep learning based binary and semantic change detection in ultra high resolution remote sensing imagery: From benchmark datasets to urban application," *ISPRS J. Photogramm. Remote Sens.*, vol. 193, pp. 164–186, Nov. 2022.

[7] N. Wang, W. Li, R. Tao, and Q. Du, "Graph-based block-level urban change detection using Sentinel-2 time series," *Remote Sens. Environ.*, vol. 274, Jun. 2022, Art. no. 112993.

[8] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[9] T. Liu, L. Yang, and D. Lunga, "Change detection using deep learning approach with object-based image analysis," *Remote Sens. Environ.*, vol. 256, Apr. 2021, Art. no. 112308.

[10] P. Yuan, Q. Zhao, X. Zhao, X. Wang, X. Long, and Y. Zheng, "A transformer-based Siamese network and an open optical dataset for semantic change detection of remote sensing images," *Int. J. Digit. Earth*, vol. 15, no. 1, pp. 1506–1525, Dec. 2022.

[11] K. Yang et al., "Asymmetric Siamese networks for semantic change detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609818.

[12] Z. Zheng, A. Ma, L. Zhang, and Y. Zhong, "Change is everywhere: Single-temporal supervised object change detection in remote sensing imagery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15193–15202.

[13] L. Ding et al., "A survey of sample-efficient deep learning for change detection in remote sensing: Tasks, strategies, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 3, pp. 164–189, Sep. 2025.

[14] M. Gong, J. Zhao, J. Liu, Q. Miao, and L. Jiao, "Change detection in synthetic aperture radar images based on deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 1, pp. 125–138, Jan. 2016.

[15] F. Cui and J. Jiang, "MTSCD-Net: A network based on multi-task learning for semantic change detection of bitemporal remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 118, Apr. 2023, Art. no. 103294.

[16] Y. Deng et al., "Feature-guided multitask change detection network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9667–9679, 2022.

[17] L. Wang, M. Zhang, X. Gao, and W. Shi, "Advances and challenges in deep learning-based change detection for remote sensing images: A review through various learning paradigms," *Remote Sens.*, vol. 16, no. 5, p. 804, Feb. 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.

[18] M. Stoimchev, J. Levatić, D. Kocev, and S. Džeroski, "Semi-supervised multi-label classification of land use/land cover in remote sensing images with predictive clustering trees and ensembles," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4706416.

[19] M. Zhang, G. Xu, K. Chen, M. Yan, and X. Sun, "Triplet-based semantic relation learning for aerial remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 266–270, Feb. 2019.

[20] H. Fang, S. Guo, X. Wang, S. Liu, C. Lin, and P. Du, "Automatic urban scene-level binary change detection based on a novel sample selection approach and advanced triplet neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5601518.

[21] K. Tang, F. Xu, X. Chen, Q. Dong, Y. Yuan, and J. Chen, "The ClearSCD model: Comprehensively leveraging semantics and change relationships for semantic change detection in high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 299–317, May 2024.

[22] H. Chang, P. Wang, W. Diao, G. Xu, and X. Sun, "A triple-branch hybrid attention network with bitemporal feature joint refinement for remote-sensing image semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5613816.

[23] T. Liu, J. Li, W. Cao, M. Tang, and G. Yang, "MLCNet: Multitask level-specific constraint network for building change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11823–11838, 2024.

[24] Q. Shen, S. Tao, R. Yang, X. Zhang, and M. Wang, "BSCDNet: A building change detection network with category differentiation using a graph attention mechanism and multitask learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 15761–15776, 2024.

[25] Z. Li, X. Wang, S. Fang, J. Zhao, S. Yang, and W. Li, "A decoder-focused multitask network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609115.

[26] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

[27] Z. Zhou, K. Hu, Y. Fang, and X. Rui, "SChanger: Change detection from a semantic change and spatial consistency perspective," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 10186–10203, 2025.

[28] J. Long, M. Li, X. Wang, and A. Stein, "Semantic change detection using a hierarchical semantic graph interaction network from high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 318–335, May 2024.

[29] Z. Sun, H. Song, K. Zhang, G. Dong, L. Liang, and Y. Zhao, "Segment anything model guided semantic knowledge learning for remote sensing change detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 5830–5834.

[30] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[31] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 596–608.

[32] H. Mai, R. Sun, T. Zhang, and F. Wu, "RankMatch: Exploring the better consistency regularization for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3391–3401.

[33] X. Wang, H. Bai, L. Yu, Y. Zhao, and J. Xiao, "Towards the uncharted: Density-descending feature perturbation for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3303–3312.

[34] Y. Zuo et al., "Robust instance-based semi-supervised learning change detection for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4404815.

[35] L. Hoyer, D. J. Tan, M. F. Naeem, L. V. Gool, and F. Tombari, "SemiVL: Semi-supervised semantic segmentation with vision-language guidance," in *Proc. Eur. Conf. Comput. Vis.*, 2023, pp. 257–275.

[36] K. Ahn and A. Cutkosky, "Adam with model exponential moving average is effective for nonconvex optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 94909–94933.

[37] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[38] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[39] X. Wang, R. Zhang, C. Shen, T. Kong, and L. Li, "Dense contrastive learning for self-supervised visual pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3024–3033.

[40] H. Xie et al., "PRCL: Probabilistic representation contrastive learning for semi-supervised semantic segmentation," *Int. J. Comput. Vis.*, vol. 132, no. 10, pp. 4343–4361, Oct. 2024.

[41] Y. Xin, Z. Fan, X. Qi, Y. Zhang, and X. Li, "Confidence-weighted dual-teacher networks with biased contrastive learning for semi-supervised semantic segmentation in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614416.

[42] M. Liao, S. Tian, Y. Zhang, G. Hua, W. Zou, and X. Li, "Calibration-based multi-prototype contrastive learning for domain generalization semantic segmentation in traffic scenes," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 12, pp. 20985–21001, Dec. 2024.

[43] J. Zhang, Y. Liu, H. Yang, and J. Cao, "FedTGP: Trainable global prototypes with adaptive-margin-enhanced contrastive learning for data and model heterogeneity in federated learning," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 15, pp. 16768–16776.

[44] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "iCaRL: Incremental classifier and representation learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2001–2010.

[45] C.-Y. Wu, C. Feichtenhofer, H. Fan, K. He, P. Krahenbuhl, and R. Girshick, "Long-term feature banks for detailed video understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 284–293.

[46] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5686–5696.

[47] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[48] Y. Tang, S. Feng, C. Zhao, Y. Chen, Z. Lv, and W. Sun, "A semantic change detection network based on boundary detection and task interaction for high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 9, pp. 17184–17198, Sep. 2025.

[49] S. Tian, Z. Zheng, A. Ma, and Y. Zhong, "Hi-UCD: A large-scale dataset for urban semantic change detection in remote sensing imagery," 2020, *arXiv:2011.03247*.

[50] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.

[51] R. Caye Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Multitask learning for large-scale semantic change detection," *Comput. Vis. Image Understand.*, vol. 187, Oct. 2019, Art. no. 102783.

[52] A. E. Maxwell, T. A. Warner, and L. A. Guillén, "Accuracy assessment in convolutional neural network-based deep learning remote sensing studies—Part 1: Literature review," *Remote Sens.*, vol. 13, no. 13, p. 2450, Jun. 2021.

[53] Y. Niu, H. Guo, J. Lu, L. Ding, and D. Yu, "SMNet: Symmetric multi-task network for semantic change detection in remote sensing images based on CNN and transformer," *Remote Sens.*, vol. 15, no. 4, p. 949, Feb. 2023.

[54] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, and P. He, "SCDNET: A novel convolutional network for semantic change detection in high resolution optical remote sensing imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102465.

[55] L. Ding, J. Zhang, H. Guo, K. Zhang, B. Liu, and L. Bruzzone, "Joint spatio-temporal modeling for semantic change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610814.

[56] X. Zhang et al., "Remote sensing image semantic change detection boosted by semi-supervised contrastive learning of semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5624113.



Yupeng Gao received the M.Sc. degree in computer science and technology from Inner Mongolia Agricultural University, Hohhot, China, in 2024. He is currently pursuing the Ph.D. degree with the University of Electronic Science and Technology of China (UESTC), Chengdu, China.

His research interests include multimodal large models and change detection.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.



Tianyu Li received the master's degree in control science and engineering from Shanghai Jiao Tong University, Shanghai, China, in 2013, and the master's degree in mathematics and the Ph.D. degree from the School of Electrical and Computer Engineering, Purdue University, West Lafayette, IN, USA, in 2022 and 2023, respectively.

He is currently an Assistant Researcher with the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China. His research focuses on image processing and time-series analysis using statistical and deep learning-based approaches.



Keli Wang is currently pursuing the master's degree with the University of Electronic Science and Technology of China, Chengdu, China.

Her research focuses on multimodal semantic segmentation, working on integrating visual, textual, and other sources of information to enhance semantic understanding and segmentation accuracy in complex scenes.



Guoqing Wang (Member, IEEE) received the Ph.D. degree from the University of New South Wales, Sydney, NSW, Australia, in 2021.

He is currently a Full Professor with the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China. He has authored or co-authored more than 40 scientific articles at top venues, including *International Journal of Computer Vision* (IJCV), IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, ICCV, and ACM MM. His research interests include machine learning and autonomous systems, with special emphasis on visual perception and embodied agents.

Dr. Wang received Australian Dean's Award for Outstanding Ph.D. Theses for his research work with UNSW.



Jiayi Ma (Senior Member, IEEE) received the Ph.D. degree in control science and engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2014.

He is currently a Professor with Wuhan University, Wuhan. He has authored or co-authored more than 200 refereed journal articles and conference papers, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON IMAGE PROCESSING, the *International Journal of Computer Vision* (IJCV), CVPR, ICCV, and ECCV. He was identified in the 2019–2023 highly cited researcher lists from the Web of Science Group.

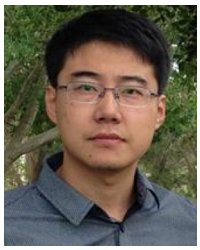
Dr. Ma is an Area Editor of *Information Fusion*.



Ningjuan Ruan received the Ph.D. degree in electronic science and technology from Nanjing University, Nanjing, China, in 2019.

She is currently the Director of the Advanced Optical Remote Sensing Beijing Key Laboratory, Beijing, China. She is also with Beijing Institute of Space Mechanics and Electricity, Beijing.

Dr. Ruan is also a member of the Expert Committee of the Core Technology Integration Platform, Nanjing University, under the Ministry of Education, and the Academic Committee of the Key Laboratory of Space Optoelectronic Detection and Perception, Ministry of Industry and Information Technology.



Yang Yang (Senior Member, IEEE) received the bachelor's degree in computer science from Jilin University, Changchun, China, in 2006, the master's degree in computer science from Peking University, Beijing, China, in 2009, and the Ph.D. degree in computer science from The University of Queensland, Brisbane, QLD, Australia, in 2012.

He is currently a Full Professor with the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China. His research interests include multimedia content analysis, computer vision, and social media analytics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 16:08:35 UTC from IEEE Xplore. Restrictions apply.